Fine-Tuning Qwen2.5-VL for Ancient Tamil Inscription

Abstract

Ancient stone inscriptions written in scripts such as Tamili and Vattezhuthu are an important source of historical and linguistic knowledge. However, reading and interpreting these inscriptions is difficult because the text is often damaged, unevenly carved on stone surfaces, and written in scripts that are no longer commonly used. In addition, there are very limited labeled datasets available for training modern machine learning models. This project aims to explore how recent vision–language models can be used to automatically recognize and interpret these inscriptions. The proposed system combines transformer-based vision language model Qwen 2.5-7B to extract text from images of inscriptions. Because real inscription datasets are limited, synthetic images will be generated by rendering text and placing it on stone-like backgrounds. To ensure realism, we have utilized the “Aksharamukha” python package to convert the inscription text to image and used "Augraphy" library along with standard pre-processing to combine the inscription image to stone background. After this, qwen 2.5-7B model is then fine-tuned using a parameter-efficient method LoRA on our custom dataset and qwen2.5-0.5B model is used to convert the recognized characters to human recognizable form. After recognizing the characters and words, additional language processing methods will be used to reconstruct sentences and convert the text into modern Tamil. The system will be evaluated using BLUE and ROUGE evaluation metrics. The overall goal of this work is to investigate how modern NLP techniques can support the digital preservation and interpretation of historical inscriptions.

1 Introduction

Historical Stone inscriptions particularly those written in archaic scripts like Vattezhuthu and Tamili serve as the primary source of socio-political and linguistic evolution of South Asia. These epigraphic artifacts are not merely relics, they are foundational to understanding the transition of languages like Tamil, which is one of the oldest language in the world, from their ancient roots to modern forms. However, archaeologists face challenges such as physical degradation of stone inscriptions due to centuries of erosion and a declining number of paleography experts capable of deciphering these inscriptions.

Recent advances in Artificial Intelligence, particularly in the field of Optical Character Recognition (OCR), have achieved near human-parity for ancient scripts, but it suffers significantly when applied to epigraphic data. The inconsistent carving depth, the irregular topography of the stone, and the lack of standardized character shapes in handwritten ancient scripts render traditional computer vision methods ineffective.

To tackle these challenges, this research investigates the use of Vision–Language Models (VLMs) for the automatic recognition and translation of ancient inscriptions. In this work, we propose a pipeline based on the Qwen 2.5-7B architecture, which is a transformer-based model capable of processing both visual and textual information. One of the major challenges in this task is the limited availability of labeled datasets for ancient inscriptions. To address this issue, a synthetic dataset generation framework is developed. The inscription text is first rendered using the "aksharamukha" python package, which converts the modern Tamil script into its ancient predecessors, namely Tamili and Vatteluttu. In addition, the Augraphy library is used to simulate realistic stone textures, lighting variations, and surface degradation. This process helps generate a training dataset that closely resembles real-world epigraphic images.

The main part of our methodology focuses on fine-tuning the Qwen model using Parameter-Efficient Fine-Tuning (PEFT) with the LoRA (Low-Rank Adaptation) technique. This method allows the model to learn the characteristics of Tamili and Vattezhuthu scripts without requiring the large computational cost of updating all model parameters and we use qwen2.5-0.5B model to convert the learned words to human recognizable form. Here, we have used the "Mixture of Experts" concept introduced in Unit 03. In addition to recognizing characters, the system also includes a linguistic reconstruction step that helps convert the recognized text into meaningful sentences and translate them into modern Tamil. To evaluate the performance of the proposed system, BLEU and ROUGE metrics are used to measure the quality of the generated

text. These evaluation results provide a quantitative way to assess how well vision–language models can be applied to the study of ancient inscriptions. Overall, this research aims to support the digital preservation of cultural heritage by connecting the study of ancient inscriptions with modern Natural Language Processing (NLP) techniques.

2 Dataset

In order to finetune qwen 2.5-7B model and qwen 2.5-0.5B model, we required a high quality and domain specific dataset. Since, there is no publicly available dataset, our first task for this project was to create a high quality dataset. Our focus was to create a paired image-text data such that we have an image and inscription text (for finetuning qwen 2.5-7B model) along with the original tamil text (for finetuning the qwen 2.5-0.5B model)

2.1 Data Collection and Chunking

The foundational Tamil language text data was sourced from open-source dataset called "Project Madurai" which provided us with Tamil language's unicode UTF-8 text format. To ensure the text was suitable for both visual rendering and sequence modeling, the raw continuous text strings were pre-processed and segmented into meaningful and uniform chunks. Furthermore, we utilized sliding window technique to generate appropriately sized sequence lengths which later optimized the physical footprint of the text for the subsequent image generation phase.

2.2 Script Conversion and Rendering Pipeline

Next, we developed a custom typographical pipeline to automate the conversion of the modern Tamil text chunks (which we obtained in the previous step) to their ancient script equivalent. Furthermore, this pipeline was also responsible to utilize "Aksharamukha" python library to convert the chunks to text images.

2.3 Synthetic Degradation and Background Blending

Now, our task is to combine the text image and blend it with a particular stone background. We used some sample open-source images as background images. We utilized "Pillow" python library to combine the text image to stone background and later used "Augraphy" pythyon library to add some degradation to the image. Specific hyper parameters were applied within Augraphy library to simulate realistic historical weathering, including:

- **Surface Noise**: Added noise to imitate erosion and small pits in the stone surface.
- **Lighting Variance**: Introduced lighting variations to recreate the uneven shadows caused by carved surfaces.
- **Carving Bleed**: Slightly modified stroke thickness to mimic irregularities that occur during manual chiseling.

2.4 Dataset Preparation

The final output of the pipeline produced a dataset of approximately 28,000 paired text–image samples. To match the input format required by the *Qwen2.5-VL* architecture, the ground-truth data was converted into a conversational JSONL structure. This format explicitly links each simulated stone image with its corresponding modern Tamil transcription.

Due to hardware limitations during training specifically the 20GB VRAM limit on the available computing cluster, it was not possible to train directly on the full 28,000-image dataset. Running the entire dataset would likely cause Out-Of-Memory (OOM) errors and VRAM fragmentation during gradient computation. To address this issue, the dataset was stratified and reduced to a representative subset of 5,000 image–text pairs, which was used for the main training process.

2.5 Sample output

A representative example of the final synthetic dataset, demonstrating the successful blending of the rendered ancient script with the physical stone textures, is illustrated below.

Figure 1: Vattelethzu image

3 Methodology

In order to deal with the challenges related to visual degradation in physical stone inscriptions

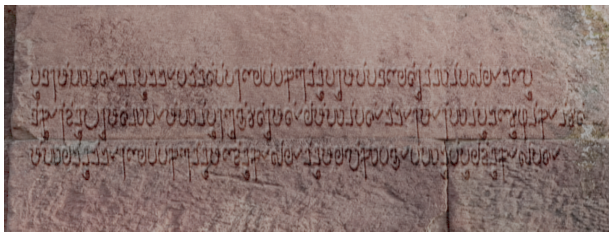

Figure 2: Tamili Image

and morphological complexities of continuous ancient Tamil scripts, our research proposes two stage multi-modal architectures. Our research isolates the character recognition task from the character segmentation task which means that character recognition task is performed by qwen 2.5-7B model and character segmentation task is performed by qwen 2.5-0.5 B model. This drastically reduces computational overhead and optimizes the mathematical convergence of both models.

3.1 The Train-Test Split

Before initializing the training loop, our dataset required rigorous preprocessing and reduced sample size to accommodate the hardware limitations, specifically a strict 20GB VRAM ceiling on the available computing node and Out of Memory errors.

Next, a stratified sample of exactly 5,000 image-text pairs was isolated for the primary fine-tuning of the vision-language model, ensuring a balanced distribution between the Tamili and Vatteluttu scripts. From this dataset of 5,000 image-text pairs, we applied a train-test split of 80/20. This yielded 4,000 images for training purpose and the rest 1,000 images for validation and zero-shot evaluation.

For the textual segmentation phase, the corresponding text transcripts were processed using a sliding-window chunking algorithm, generating 25,611 sequence pairs (un-spaced to spaced), from which 1,000 were strictly reserved for validation.

3.2 Model Architecture

The project utilizes two distinct foundation models from the qwen 2.5 family. One for extracting text from the image and another for segmenting the output text received from the previous model to human readable form.

3.2.1 The Vision-Language Model (Qwen2.5-VL-7B)

The primary OCR process is handles by the Qwen 2.5-7B model which contains a dynamic-resolution Vision Transformer (ViT) encoder with a robust autoregressive LLM decoder. The major advantage of using this model for epigraphical research is its native spatial scale awareness that is, it utilizes Multimodal Rotary Positional Encoding (MRoPE) to directly process pixel-level coordinates without normalization, enabling highly precise spatial grounding on the complex topological surfaces of stone carvings.

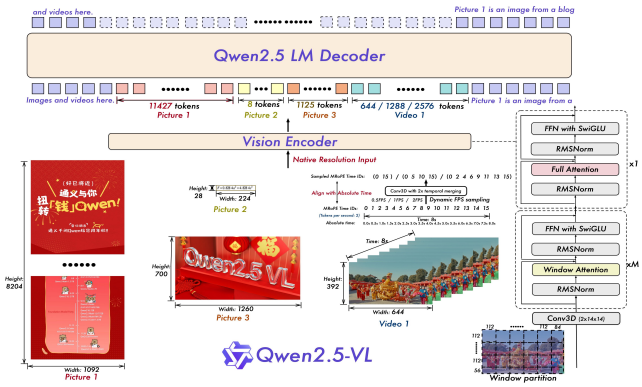

Figure 3: Tamili Image

3.2.2 The Linguistic Segmenter (Qwen2.5-0.5B)

After receiving the results from Qwen 2.5-7B model, we pass the output to Qwen 2.5-0.5B model so that it can add the required spacing among different text. The Qwen2.5-0.5B architecture consists of a 24-layer transformer decoder that is uniquely optimized for edge deployment and fast sequence-to-sequence token generation, making it an ideal, lightweight "grammarian" to post-process the raw output from Stage 1.

4 Evaluation

5 Results

6 Future work

Although, our current project achieved great results in recognizing and segmenting the text for ancient Tamil scripts like Vattezhuthu and Tamili, several avenues for further research and optimization remain. These future directions aim to improve both the linguistic accuracy and the computational cohesion of the system.

6.1 Incremental Fine-Tuning and Continual Learning

The first work we want to do is to explore the Incremental-Finetuning strategies. Like, we were

forced to reduce our dataset size from 28000 image-text pair to only 5000 images-text pair so, it will be great to see how a larger dataset will impact both model's performances.

6.2 Model Consolidation: Monolithic Multi-Tasking

The current research purposefully decouples vision and text tasks to manage hardware constraints. However, as GPU memory architectures evolve, a significant future step involves consolidating the pipeline into a single-stage multi-task model. By training a unified transformer to map raw visual features directly to segmented, human-readable modern Tamil, the system could benefit from "joint-feature learning," where the model's understanding of linguistic boundaries informs its visual character recognition, potentially reducing the accumulation of errors between stages.

6.3 Script and Language Expansion

The synthetic generation and fine-tuning framework developed in this study is script-agnostic, meaning that it can potentially be applied to other historical writing systems as well. One important direction for future work is to expand the dataset to include additional Brahmic and Dravidian scripts such as Grantha, Brahmi, and early Kannada.

Expanding the dataset in this approach could make it possible to develop a more general epigraphic transcription system capable of working with a wider range of South Asian historical inscriptions. Such a system would help researchers study and interpret historical records from different regions and time periods, supporting broader archaeological and linguistic research.

6.4 Scaling to Ultra-Large Foundation Models

With the development of newer vision–language architectures such as Qwen3-VL-235B-A22B-Instruct, there is an opportunity to explore the use of much larger models for this task. These models use Mixture-of-Experts (MoE) mechanisms and contain significantly more parameters than the 7B model used in this study.

Increasing the model scale may improve the system's ability to handle more challenging cases, such as inscriptions that are heavily eroded or partially damaged, where the current model may struggle to correctly interpret the characters.

Future work will explore the use of these ultra-large models along with techniques such as model sharding and quantization. These methods would allow the large models to be deployed more efficiently while evaluating whether increased model scale can improve performance in ancient script transcription tasks.

References

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